# A Simulation Model and Dashboard For Predicting COVID-19 Bed Requirements

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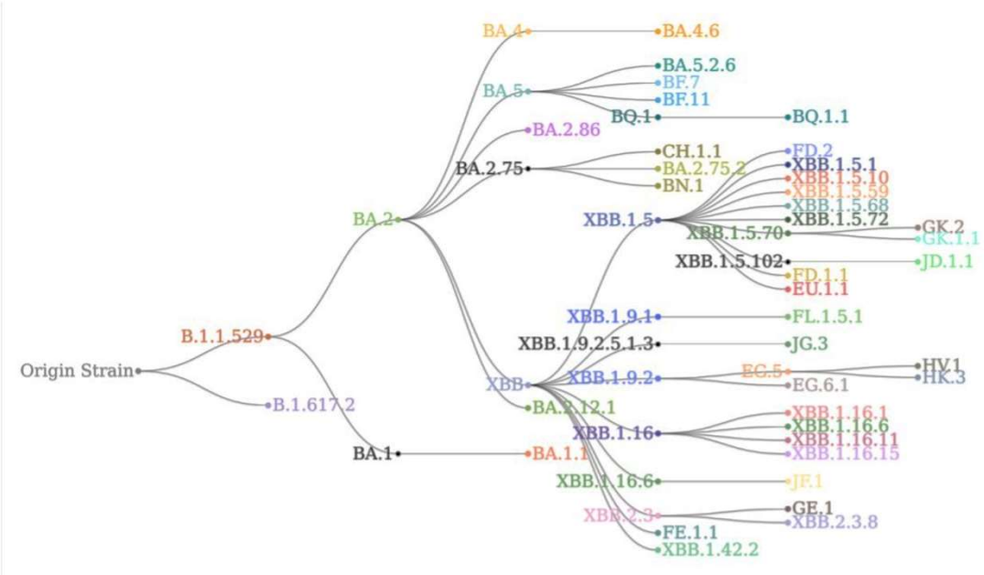
# Background: COVID-19




COVID-19 continues to evolve

Image source: https://covid.cdc.gov/covid-data-tracker/#variant-proportions

COVID-19 in the UK:

- 22.2 million total cases (as of 19 May 2022)
- 232,837 deaths (as of 27 Oct 2023)
- Over 6% prevalence in March 2022 (1 in 16)

Sources:

- https://coronavirus.data.gov.uk/
- Science, 2022. https://doi.org/10.1126/science.abq4411

"…the world failed this test of preparedness and response. It remains to be determined whether the world will do better next time—because it is inevitable that there will be a next time."

Thomas R Frieden, Marine Buissonnière, Amanda McClelland; BMJ Global Health 66(3), 2021. https://doi.org/10.1136/bmjgh-2021-005184

# Background: Healthcare in the UK

- National Health Service founded in 1948
  - "The new health service in all its branches will be free to all" [*A National Health Service – The White Paper Proposals in Brief*, HM Stationery Office (1944)]
  - In contrast: 16.6% of Texans were uninsured as of 2022 (worst in USA) KFF: https://www.kff.org/other/state-indicator/total-population/
- However, in recent years the NHS has come increasingly under strain
- Therefore:
  - Need to optimise the usage of limited medical & staff resources
  - Need contingency plans to deal with unexpected shocks in healthcare demand (e.g. due to pandemics)

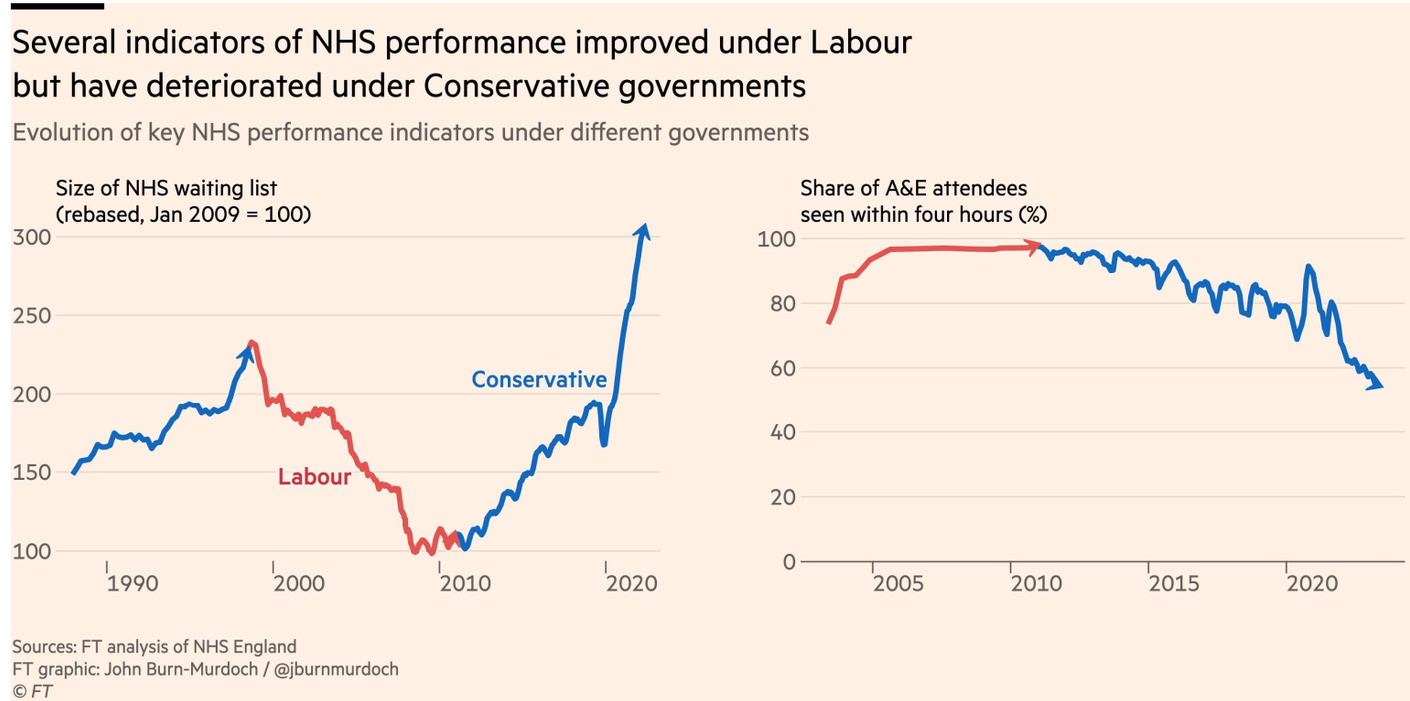


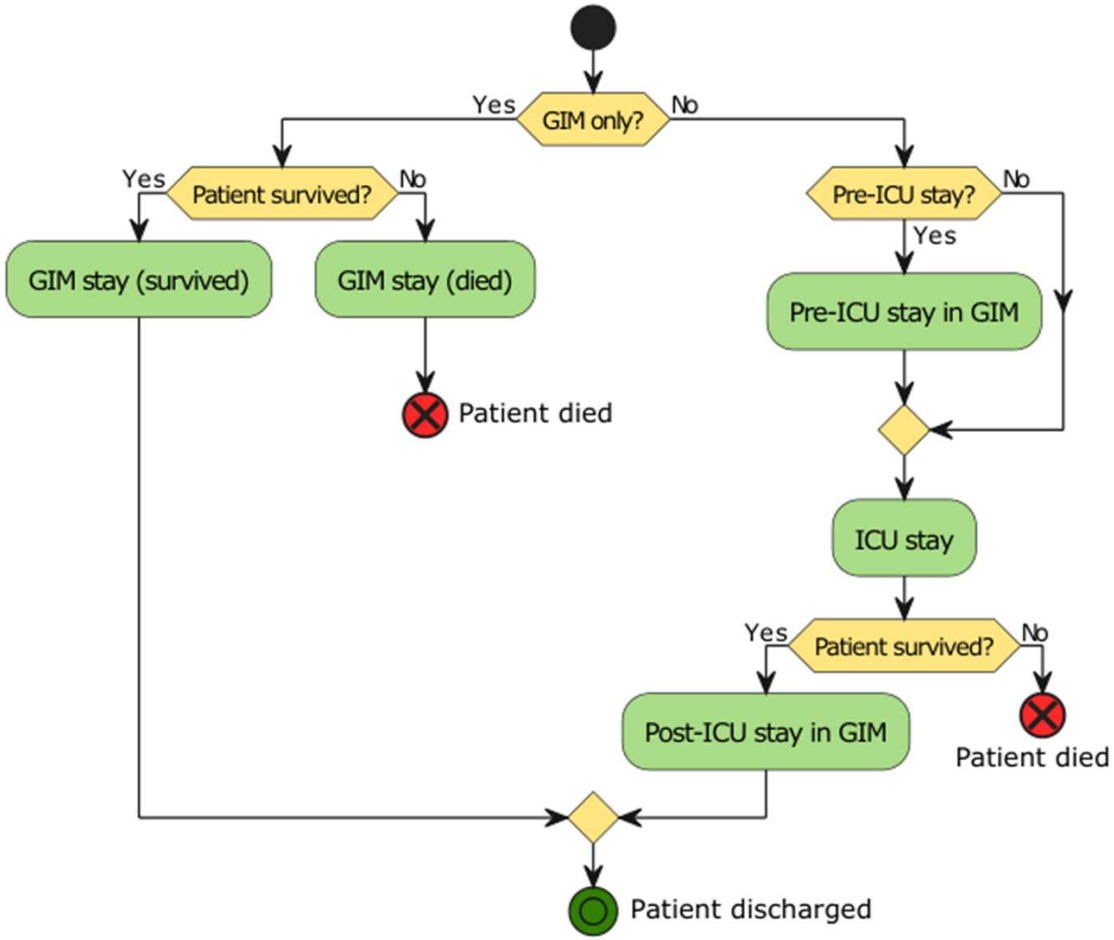

Image Source: John Burn-Murdoch / Financial Times
https://twitter.com/jburnmurdoch/status/1606223967903260673

# Case study: Addenbrooke's Hospital, Cambridge, UK

- Part of Cambridge University Hospitals NHS Foundation Trust (CUH)
- Previously: patient flow model of COVID-19 patients
  - GJ Melman, AK Parlikad, EAB Cameron; Health Care Management Science 24(2), pp. 356-374, 2021. https://doi.org/10.1007/s10729-021-09548-2
- This work:
  - Analysis of General Internal Medicine (GIM) and Intensive Care Unit (ICU) bed demand, patient lengths-of stay (LoS)
  - Discrete-event simulation model of patient flow (Python )
    - Unite data preprocessing, modelling and simulation under a single framework
    - Web dashboard implemented using Plotly Dash plotly
  - Simulation results used for "red/green" bed allocation planning at Addenbrooke's

# Patient flow model

- Five types of length-of-stay (green boxes):
  - GIM-only, patient survived
  - GIM-only, patient died
  - Pre-ICU stay in GIM
  - ICU stay
  - Post-ICU stay in GIM



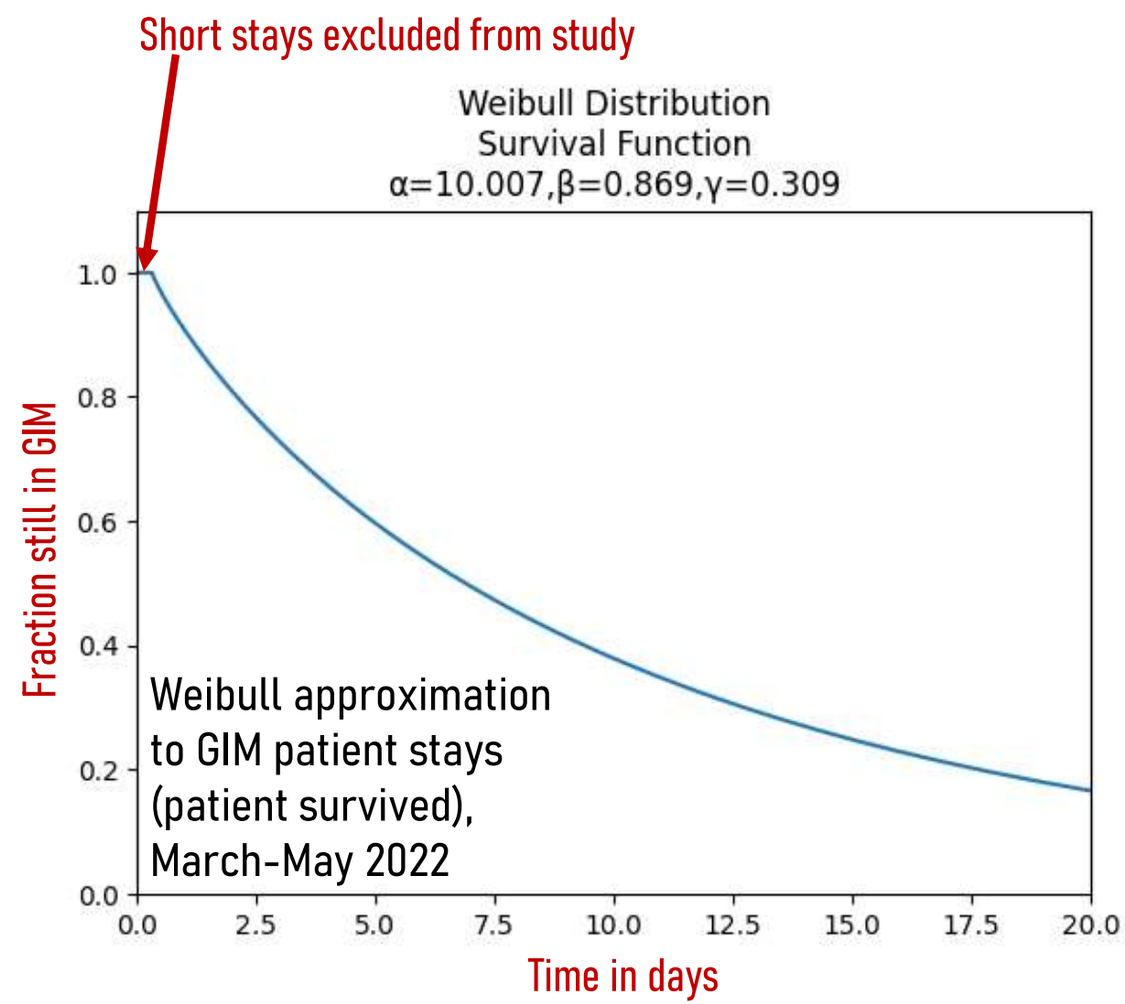
# Data Collection & Preprocessing

- Patient data mined from CUH's EPIC electronic health record system
  - Demographic data
  - Admission/discharge times for general and/or ICU ward
  - Readmission data if applicable
    - Only the first readmission was tracked in this study
  - Infection source (community / hospital)
- Preprocessing:
  - Remove short stays
  - Remove patient history prior to infection (if hospital-acquired)
  - Combine stays if gap between first admission and readmission is small

# Length-of-Stay Analysis (1)

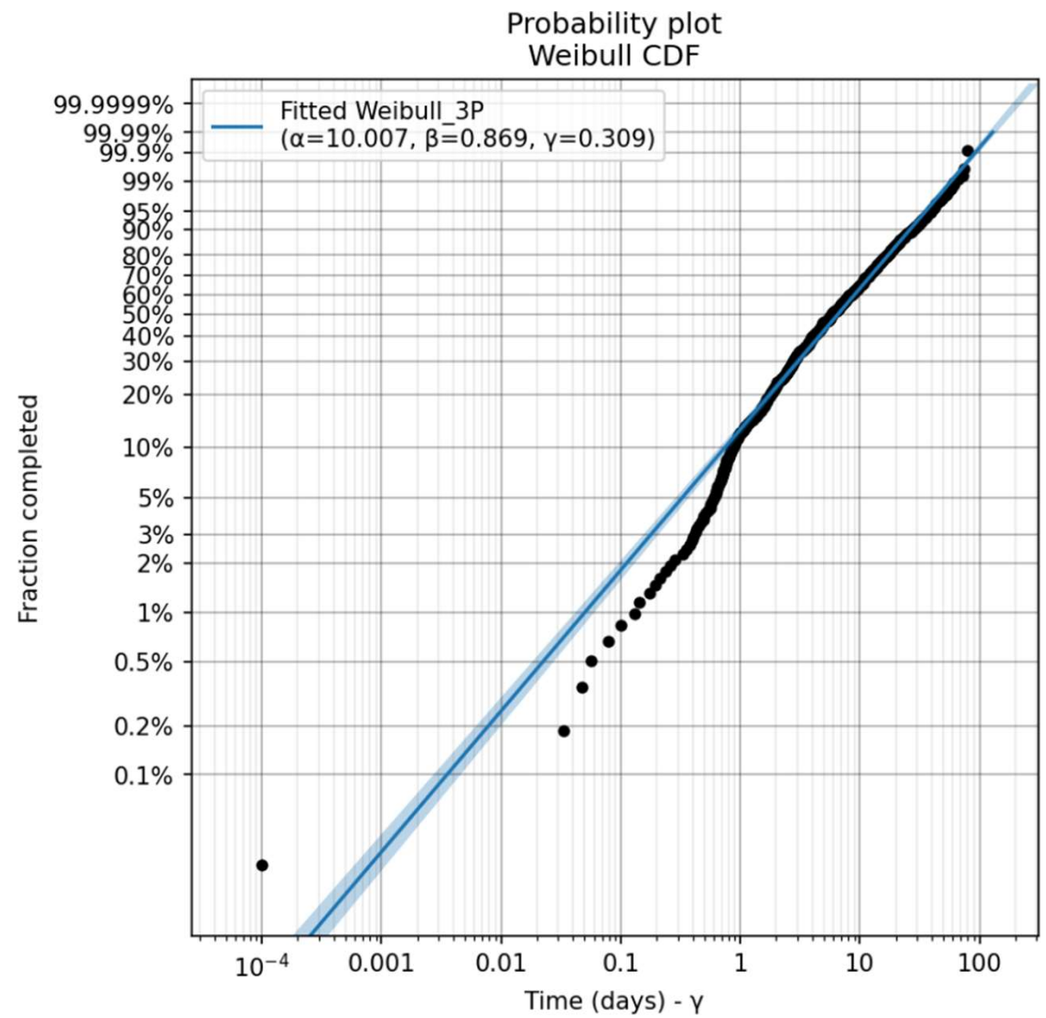
- The reliability Python package was used for distribution fitting
- A shifted Weibull distribution (`Weibull_3P`) was found to fit all five LoS types well
  - Proportion of patients remaining after $t$ days:

$$\bar{F}(t) = \exp\left[-\left(\frac{t-\gamma}{\alpha}\right)^{\beta}\right] \qquad (t \geq \gamma)$$



# Length-of-Stay Analysis (2)

- Probability plots were used to visualise the goodness of fit
  - In general, fit is poorer for shorter stays, but such stays have less effect on overall bed usage
  - Shifted Lognormal and Gamma distributions were also considered, with similar results

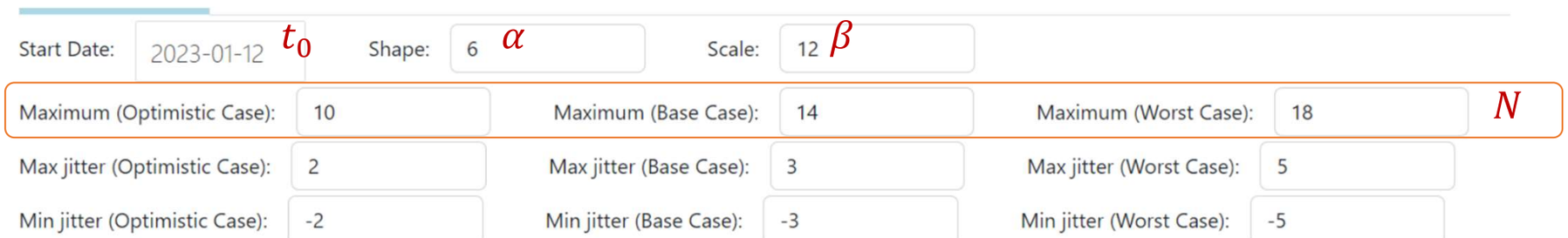


# Step-down policy

- Long-stay patients generally are not infectious after a certain amount of time
  - Typically admitted for other reasons
  - "With" vs. "for" COVID-19
- Step-down policy introduced to reflect change in patient status
  - GIM patients removed from "red bed" status after $N$ days (if still in hospital)
  - In reality, patients are only removed after negative test results
    - Would require data on inpatient testing results – not done in this study
- Our simulation model considers both total and "red" bed usage by patients admitted with COVID-19

# COVID-19 Patient Arrival Modelling (1)

- Gamma functions were used to model daily patient arrivals
  - Parameters $\alpha, \beta, N, t_0$
  - Maximum value at $t = t_0 + (\alpha - 1)\beta$ such that $f(t) = N$
- For prediction, we would create 3 scenarios, varying the maximum value $N$
- A visual tool was developed for parameter selection



# COVID-19 Patient Arrival Modelling (2)

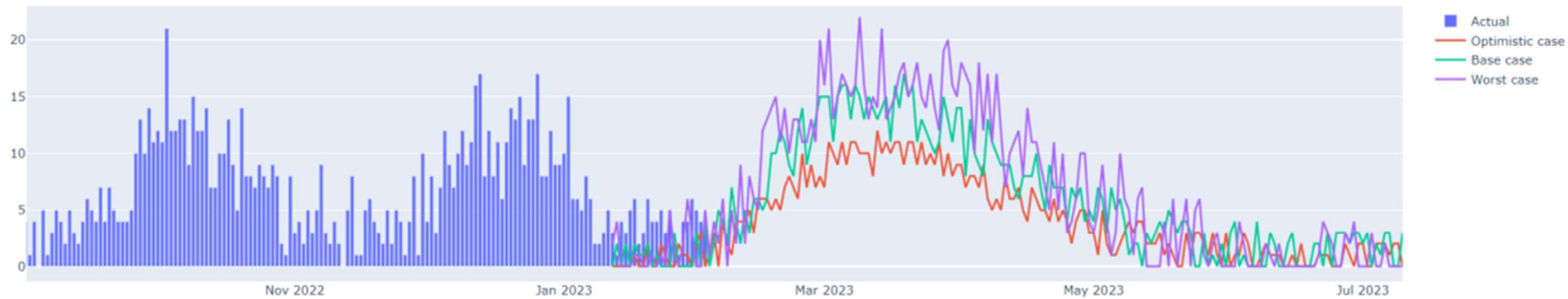
Modelling Tool | Info

Start Date: 2023-01-12 Shape: 6 Scale: 12

Maximum (Optimistic Case): 10 Maximum (Base Case): 14 Maximum (Worst Case): 18

Max jitter (Optimistic Case): 2 Max jitter (Base Case): 3 Max jitter (Worst Case): 5

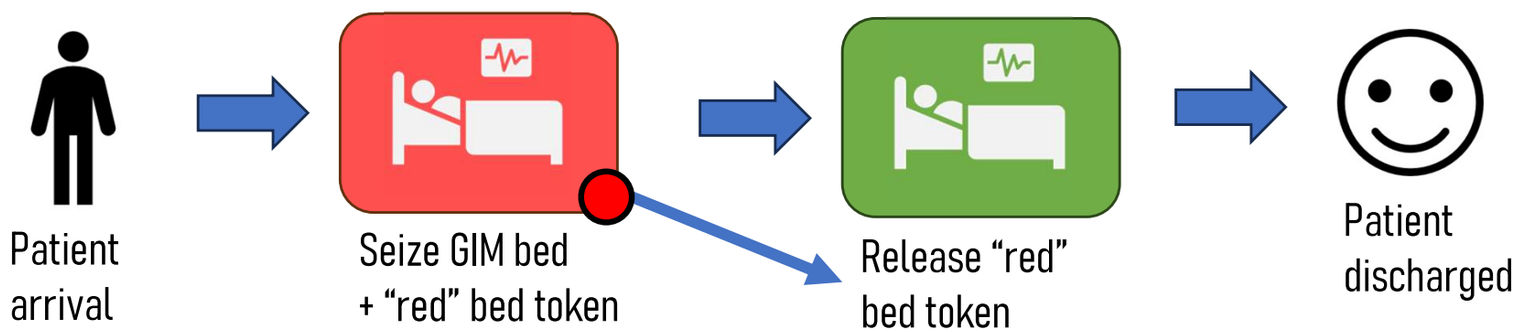
Min jitter (Optimistic Case): -2 Min jitter (Base Case): -3 Min jitter (Worst Case): -5




Scenario Start Date: 2022-09-01 Download file

# Python Simulation

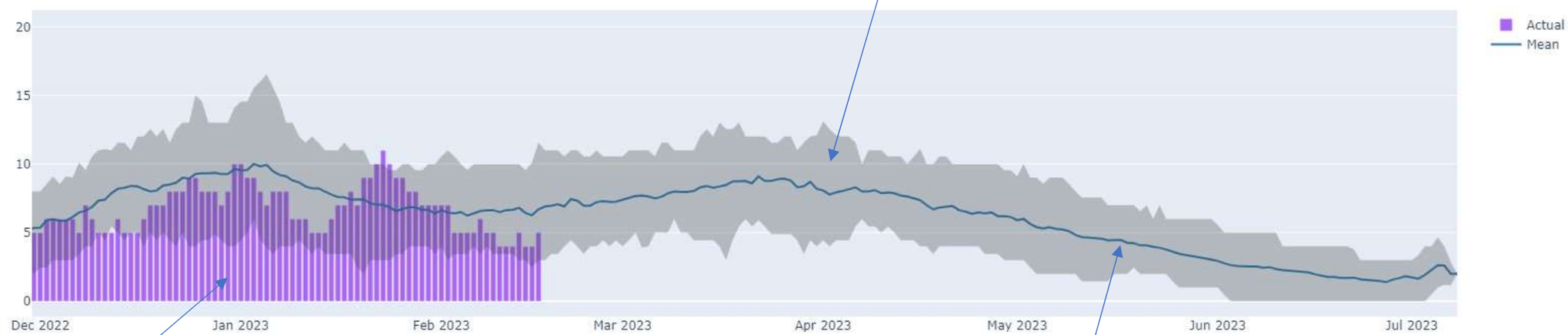
- Discrete event simulation using simpy
  - Based on Python generators
- Bed occupancy logged using pandas dataframes
- Trick to track "red" bed status:
  - Patients in GIM seize both an actual GIM bed and a "red bed" token
  - Releasing the token = transfer to "green" bed status (assuming patient is not immediately discharged)



# Example simulation output

90% credible interval
(5th and 95th percentiles)

ICU bed occupancy

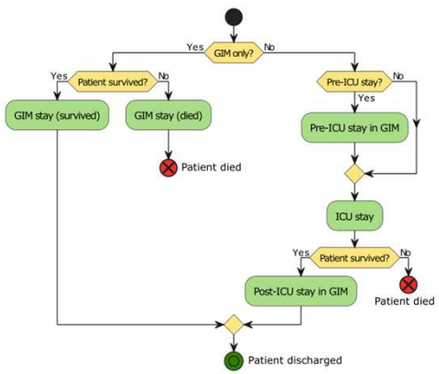



Actual bed occupancy

Mean estimated bed occupancy

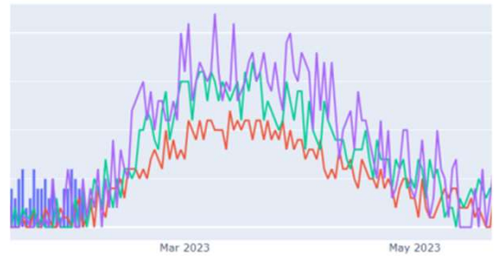
# Dashboards

- Plotly Dash used to create dashboards to simplify modelling and simulation
  - Hybrid open-source/commercial product – we only use open-source components
- Three dashboards



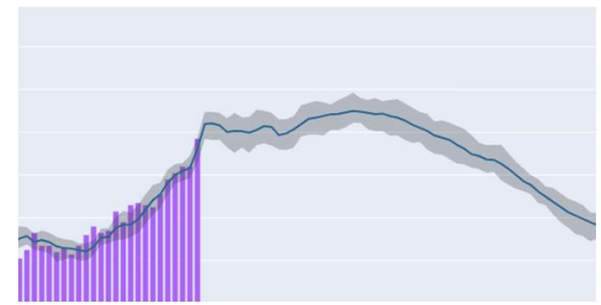

<u>Patient analysis</u>

Fit length-of-stay distributions and compute probability of each patient pathway




<u>Scenario creation</u>

Generate number of daily patient arrivals to the simulation model



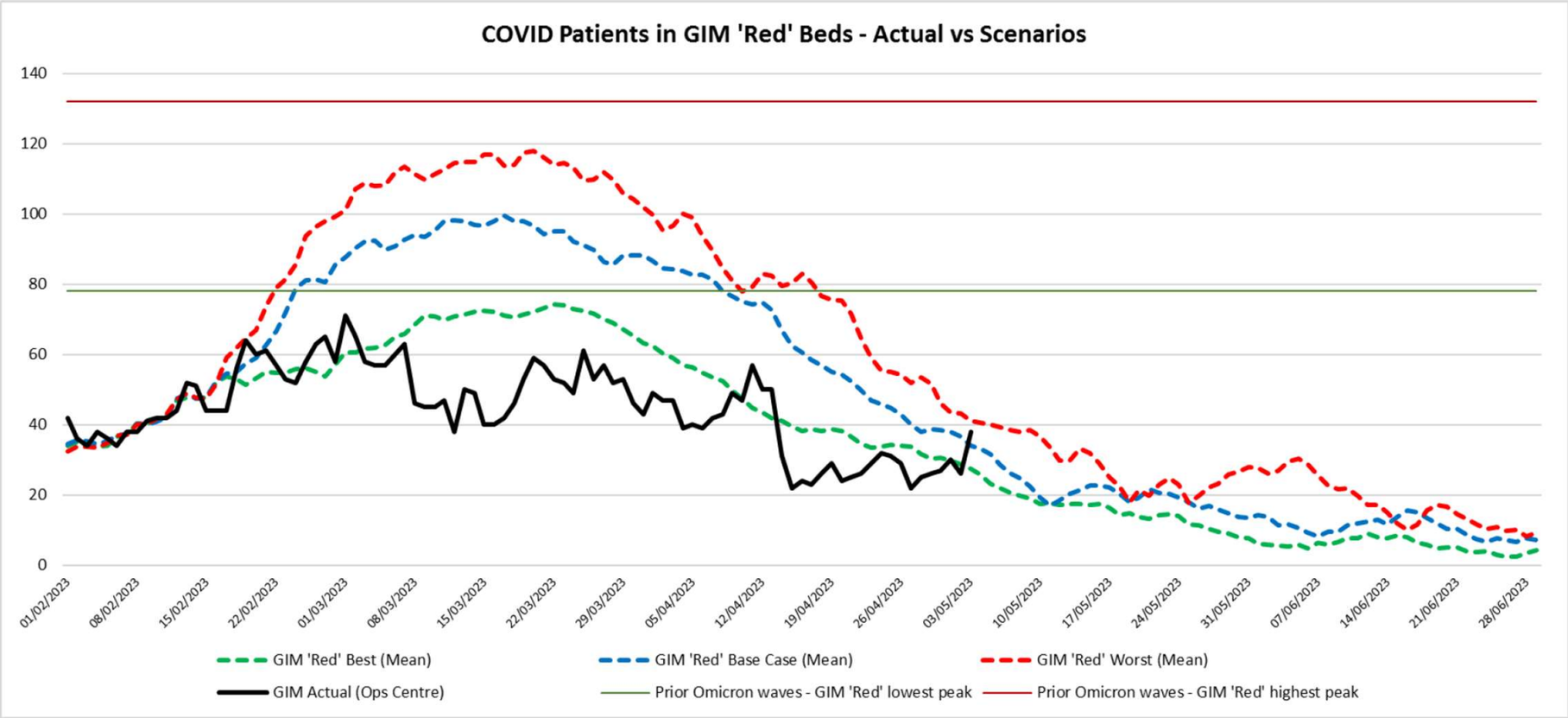
<u>Simulation</u>

Simulate the model with the given parameters and plot the results

# Dashboard Usage

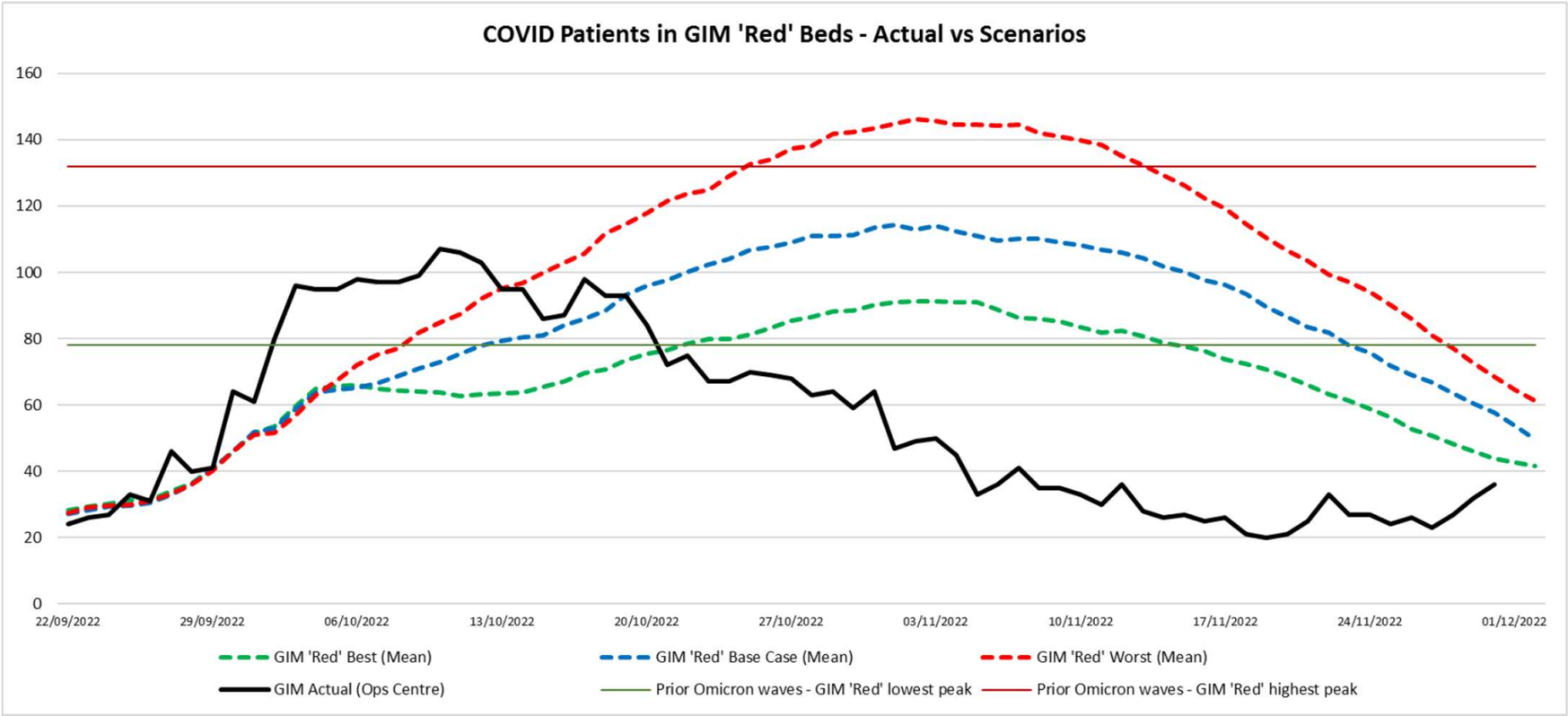
- Used to make medium-term (around 3 months) predictions on bed demand at Addenbrooke's Hospital
  - Complimented by an Early Warning Tool, which focused on short-term forecasting (up to 3 weeks)
- Typically, three scenarios (optimistic, base, "worst-case") would be generated near the start of a COVID-19 wave
  - Predict peak daily admissions and wave duration
  - Patient length-of-stay and ICU admission rate/survivability estimated using the previous wave
- We would generally run 30 simulation replications to generate credible intervals, but only report the means (see next slides)

# Example: COVID-19 wave starting around Feb 2023




- "Red" bed occupancy close to (but slightly below) projected "best-case" scenario

# Example: COVID-19 wave starting around Sept 2022




- Bed occupancy peak turned out to be much earlier than projected, with quick decline
- Demonstrates difficultly of making accurate predictions, especially early in a wave
- Long-term predictions should be complimented with short-term "nowcasting"

# Looking Forward

- Currently examining applying the model to flu/RSV/other respiratory illnesses
- Future Technical improvements?
  - Automatic curve fitting for previous disease waves
  - Random noise in daily arrivals should be generated separately for each simulation replication (currently all replications use the same daily arrival numbers)
  - Ability to add baseline levels of patient admissions (more flexibility in scenario modelling)
    - Statistics from recent months show consistent low level of COVID-19 admissions

# Summary

- A Python simulation program was used to estimate bed occupancy by COVID-19 patients at Addenbrooke's Hospital, Cambridge, UK
- Estimates were used to plan "red/green" bed allocation (red = infection control)
- A Python dashboard was created for easy data analysis + scenario creation + simulation
- Currently planning to use dashboard for flu/RSV analysis

# Thank you!

# Extra Content

The details…

# Brief overview of SimPy

- Main classes used from SimPy:
  - Environment: the simulation state and event handling loop
    - Processes are added to the simulation using Environment.process()
  - Resource: seized/released by processes
    - Since we want to know hypothetical bed occupancy under no capacity limits, we set all our resources to have infinite capacity
- Processes are Python generators
  - yield keyword passes control back to the event handling loop – e.g. when waiting to seize a resource
  - Can nest generators using yield from → create subprocesses

# Main patient arrival generator

```python
def arrival_generator(env: Environment):
    """Generates patients for the simulation `Environment`.

    Args:
        env (Environment): The simulation `Environment`.

    Yields:
        simpy.events.Process: Each new patient's `Process`.
    """

    # Note that new arrivals are generated at midnight each day.
    # The `new_arrival` process is responsible for setting the
    # actual admission time of each patient within the day.
    # All time units in the Environment are in days.

    for n_arrivals in env.daily_arrivals:  # for each day
        for _ in range(n_arrivals):  # for each arrival in the day
            env.process(new_arrival(env))
        yield env.timeout(1.)  # Wait until next day
```

- Number of arrivals for each day pre-generated in an Excel file
- All arrivals for each day are generated at midnight
  - The new_arrival generator holds each patient for a random amount of time
  - Arrivals are thus randomly distributed throughout the day

# Launching each patient's process generator

```python
def new_arrival(env: Environment):
    """Generator process for a new patient arrival.

    Args:
        env (Environment): The simulation environment.

    Yields:
        A simpy Event.
    """
    # Get the time of day of the arrival (as a random number between 0 and 1)
    arrival_offset = env.hourly_dist.rvs() / 24.0 + uniform(0.0, 1.0 / 24.0)
    yield env.timeout(arrival_offset)

    if random() < env.params.icu_rate:
        yield from icu_process(env)
    else:
        yield from gim_process(env)
```

Probability of a patient arriving within each hour is taken from historical data

Delays in SimPy are done by waiting for a timeout event

Chooses which subprocess to follow based on a probability

# Brief overview of Dash

- Components are Python classes
  - Anything from a simple HTML element to an entire plot figure
  - Components can be assigned an ID (string), child components, and other properties
- Interactivity is added using callbacks

```
@app.callback(
  Input('component1', 'property1'),
  Input('component1', 'property2'),
  Output('component2', 'property3'),
  Output('component2', 'property4'),
  State('component3', 'property5')
)
def my_callback(prop1, prop2, prop5):
  #code goes here
  return prop3, prop4
```

In this example, changes to property1 or property2 trigger this callback, which changes the values of property3 and property4. The value of property5 is read, but changes to it do **not** trigger the callback.